

Binary Logistic Regression

Jinsong Chen

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Chapter overview

A yes-or-no outcome has a mean that is a probability. This chapter builds a model for that probability, connects it to odds and log-odds, and explains why the same odds ratio can imply very different probability differences. A known-process certification example makes the model's behavior visible; a study of student volunteering supplies the empirical analysis. Maximum likelihood, model comparisons and diagnostic displays lead to a report whose event, scale, uncertainty and scope are explicit. The appendices extend numerical fitting and simulation without supplying a prerequisite for the main analysis.

Cowles and Davis recorded personality scores and volunteering for an experiment among 1,421 introductory psychology students (Cowles & Davis, 1987). There were 597 volunteers and 824 non-volunteers. The question is how the probability of volunteering differs across extraversion scores, and how that comparison changes after allowing for recorded sex and neuroticism. A department interested in recruitment needs an answer on a usable scale: a detectable association need not separate future volunteers from other students well enough to guide a decision.

The distributed records are `carData::Cowles` (Fox et al., 2022; Fox & Weisberg, 2019). They contain one row per student, with no missing values in these fields. The extract supplies no probability-sampling or classroom-clustering information. The fitted comparisons therefore describe these recorded students under a working independent-row model; generalization and causal interpretation require additional design information.

Displayed results are rounded to four decimal places, with trailing zeros dropped unless they align a comparison. Calculations using different rounding or software settings may differ slightly.

1 A Binary Outcome and Its Probability

1.1 Define the event and the comparison

A binary outcome has two categories. A binary model treats one as the **event**, coded 1, and the other as the **non-event**, coded 0. Naming the event determines what every coefficient and predicted probability means. Here the event is volunteering, and the **prevalence**, the proportion of records that are events, is .4201. The recorded non-event means only that the student did not volunteer; it does not tell us whether the student declined, was unavailable, or never received the invitation. Those distinctions concern measurement, before any statistical fit.

```
d0 <- carData::Cowles
dat <- d0
dat$event <- as.integer(dat$volunteer == "yes")
```

```
table(dat$sex, dat$event)
mean(dat$event)
```

Table 1

One observed table supports more than one scale of comparison.

Recorded sex	Volunteers	Non-volunteers	Proportion volunteering	Odds of volunteering
Female	349	431	.4474	.8097
Male	248	393	.3869	.6310

The female-minus-male probability difference is .0605; the ratio of their odds is 1.2832. Both describe the same table, but they answer different questions. The first compares proportions directly. The second compares events relative to non-events in each group. Neither adjusts for the two personality scores or identifies a consequence of changing a person's sex.

Write $\pi(x) = P(Y = 1 \mid X = x)$. Three possible comparisons of units one predictor point apart are

$$\pi(x+1) - \pi(x), \quad \frac{\pi(x+1)}{\pi(x)}, \quad \frac{\pi(x+1)/\{1 - \pi(x+1)\}}{\pi(x)/\{1 - \pi(x)\}}.$$

They are the probability difference, risk ratio and odds ratio. A model with a constant logistic slope holds the third constant across x ; the other two generally vary. An estimand for a binary outcome must name its scale as well as the units and comparison.

1.2 A probability model and its link

Applying least squares to a 0/1 outcome gives a **linear probability model**. Its slope is already on a probability-difference scale, which can be useful over a limited range. Its weakness as a general probability model is visible before fitting: an unrestricted line can leave $[0, 1]$, and one fixed probability difference cannot remain plausible near both boundaries. Moreover a binary outcome has variance $\pi(1 - \pi)$, so its conditional variance changes with its mean. Ordinary constant-variance standard errors are inappropriate; robust uncertainty can address that variance calculation but cannot keep a straight mean function in range. Normal-error small-sample t theory does not apply to two-valued errors.

The logistic alternative declares the outcome model and mean function separately:

$$Y_i \mid \mathbf{x}_i \sim \text{Bernoulli}(\pi_i), \quad E(Y_i \mid \mathbf{x}_i) = \pi_i, \quad \text{Var}(Y_i \mid \mathbf{x}_i) = \pi_i(1 - \pi_i),$$

$$\eta_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij}, \quad \text{logit}(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \eta_i, \quad \pi_i = \frac{1}{1 + e^{-\eta_i}}.$$

Here η_i is the **linear predictor**, an additive combination of predictors. The **link** connects the conditional mean to that predictor; it transforms the probability, not the observed 0 or 1. The inverse logit turns any finite η_i into a probability strictly between 0 and 1. It passes through .5 at $\eta_i = 0$ and is steepest there. The model is linear in log-odds and curved in probability. It has no separate residual variance parameter because the Bernoulli mean determines its variance.

A response distribution, a linear predictor and a link form the generalized linear model (GLM) framework (Nelder & Wedderburn, 1972). Changing the response and link extends this framework to other outcomes, while retaining the distinction between model scale and reporting scale.

Table 2

Only the response distribution and the link change.

Component	Chapter 11	Chapter 12, nominal or ordered	Chapter 12, counts
Outcome	Binary event	Nominal or ordered category	Count, with an exposure where needed
Response distribution	Bernoulli or binomial	Multinomial	Poisson or negative binomial
Link	Logit	Generalized or cumulative logit	Log
Exponentiated coefficient	Odds ratio	Relative-risk or cumulative odds ratio	Rate ratio

1.3 Conditions and the scope of interpretation

Table 3

Conditions of the working logistic model.

Condition	What it supplies or limits
AS1: a Bernoulli outcome with a well-defined event	The meaning of π_i and its variance; coding must represent the question
AS2: the log-odds is linear in the entered terms	The conditional comparison represented by each coefficient; curves or products require explicit terms

Condition	What it supplies or limits
AS3: independent rows given the predictors	The likelihood product and its usual standard errors; repeated and clustered rows need dependence-aware methods
AS4: predictors are measured and coded as intended	The interpretation of a comparison; measurement error can change the relationship being estimated
AS5: supported profiles, overlap and no separating predictor combination	A defensible comparison and a finite maximum-likelihood estimate
AS6: no perfect linear dependence among predictor columns	Unique coefficient coordinates; near dependence can make estimates imprecise

There is no homoskedasticity or Normal-residual condition. The mean model is checked on the logit scale, and the study design supplies information about independence. Observing both events and non-events in the dataset is necessary but does not rule out a combination of predictors that separates them perfectly. Nor must both outcomes appear at every exact continuous predictor value. Support means that the proposed profiles and contrasts are represented well enough to make their interpretation credible.

A coefficient compares units agreeing on the other included predictors. It becomes a causal effect only with an adequate design and identification argument. A further complication is **noncollapsibility**: a conditional odds ratio can differ from a marginal odds ratio even without confounding, because averaging nonlinear probabilities does not preserve an odds ratio (Agresti, 2018). Movement after adjustment alone therefore does not diagnose confounding. A standardized probability comparison names an averaging population explicitly and can make the substantive magnitude clearer.

2 Three Scales: Probability, Odds, and Log-Odds

2.1 Translate one quantity three ways

A probability compares events with all units. Odds compare events with non-events. Log-odds put that positive ratio on an additive, unbounded scale:

$$\text{odds} = \frac{\pi}{1 - \pi}, \quad \pi = \frac{\text{odds}}{1 + \text{odds}}, \quad \text{log-odds} = \log \frac{\pi}{1 - \pi}.$$

At probability .5, events and non-events are equally common: odds are 1 and log-odds are 0. At probability .75, there are three events per non-event. The three scales contain the same information; they organize comparisons differently.

```
p <- c(.05, .20, .42, .50, .75, .90)
cbind(probability = p, odds = p / (1 - p), log_odds = qlogis(p))
```

Table 4

Probability, odds, and log-odds for selected probabilities; the odds scale is multiplicative around 1 and the log-odds scale additive around 0.

Probability	Odds	Log-odds
.05	.0526	−2.9444
.20	.2500	−1.3863
.42	.7241	−.3228
.50	1.0000	.0000
.75	3.0000	1.0986
.90	9.0000	2.1972

Odds exceed probabilities for $0 < \pi < 1$, with a widening gap toward 1. Complementary probabilities have log-odds equal in magnitude and opposite in sign. These features explain why an event reversal inverts an odds ratio, and why odds and risk ratios agree approximately only when both compared probabilities are small.

2.2 Equal logit steps, unequal probability steps

The inverse-logit function is the cumulative distribution function of a standard logistic variable, evaluated at η . This mathematical connection offers a threshold-crossing intuition without changing the observed Bernoulli outcome model. A one-unit step in η has the same size everywhere on the logit scale; its probability consequence depends on the starting point.

```
plogis(c(-3, -2, 0, 1))
diff(plogis(c(-3, -2)))
diff(plogis(c(0, 1)))
```

Moving from -3 to -2 raises probability by .0718; moving from 0 to 1 raises it by .2311, 3.2191 times as much. A statement that one more unit adds a fixed number of percentage points is therefore incomplete for a logistic model. Coefficients and their uncertainty are convenient on the logit scale, multiplicative comparisons on the odds scale, and most practical decisions on the probability scale.

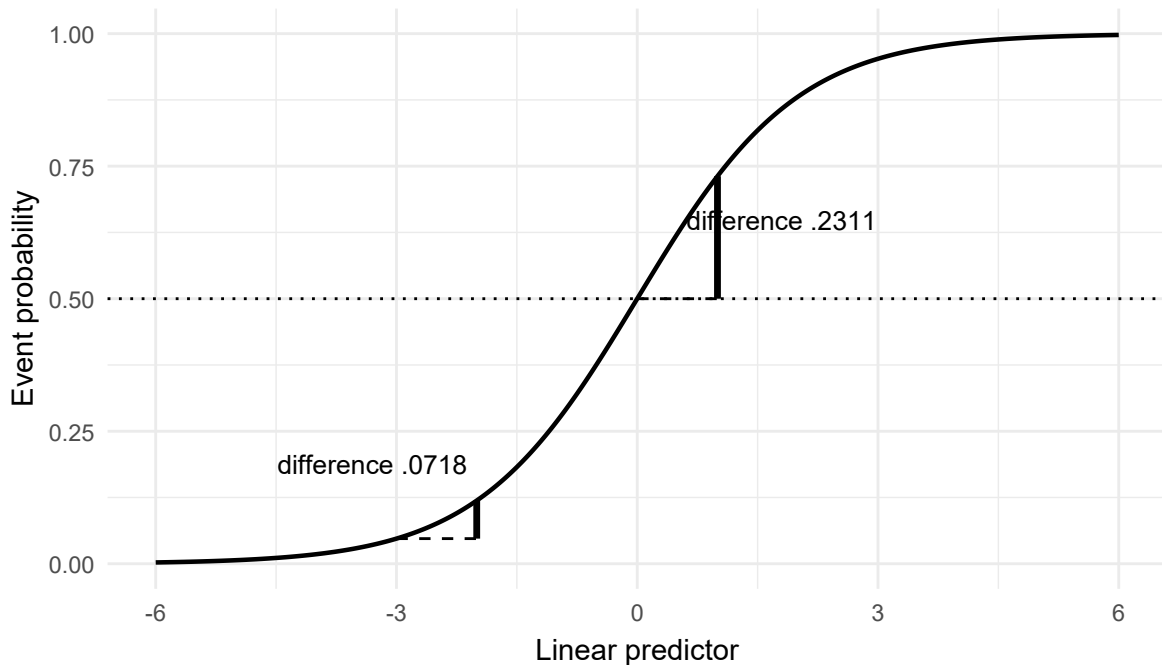


Figure 1: Equal steps on the linear-predictor scale give unequal probability differences.

3 Fit by Maximum Likelihood

3.1 From one probability to a coefficient vector

For a common probability, 32 successes in 50 fixed trials produce a binomial likelihood maximized at .64. Here each student's probability can differ through the predictor values, but the principle is the same: hold the observed data fixed and ask which parameter values make them most probable.

A small example makes the likelihood's job concrete. Three invented observations have predictor values 0, 1 and 2 and outcomes 1, 0 and 1. Compare two proposed models, A with logit $-1 + x$ and B with logit $-2 + x$. The likelihood uses the probability of what happened: p for an event and $1 - p$ for a non-event.

Table 5*Each observed outcome contributes its own probability.*

x	y	A: event	A: observed	B: event	B: observed
0	1	.2689	.2689	.1192	.1192
1	0	.5000	.5000	.2689	.7311
2	1	.7311	.7311	.5000	.5000

Predict which model has the larger product before calculating. Model A helps both events but makes the non-event less probable. Its total log-likelihood is nevertheless larger: -2.3197 against -3.1333 . Improving total fit need not improve every observation. Neither candidate has yet been claimed to maximize likelihood.

```
x <- c(0, 1, 2); y <- c(1, 0, 1)
sum(dbinom(y, 1, plogis(-1 + x), log = TRUE))
sum(dbinom(y, 1, plogis(-2 + x), log = TRUE))
```

With independent rows,

$$L(\beta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}, \quad \ell(\beta) = \sum_{i=1}^n \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\},$$

where $\pi_i = \text{logit}^{-1}(\mathbf{x}_i^T \beta)$ and $\mathbf{x}_i$ includes the intercept entry 1. Each observed event contributes its fitted probability and each non-event its fitted complement. Larger log-likelihood means better agreement for these observations. The likelihood is a function of parameters with the data fixed, not a probability distribution over those parameters.

Maximum likelihood chooses the coefficient vector maximizing this function. Its score equations are

$$\frac{\partial \ell}{\partial \beta} = \sum_i \mathbf{x}_i (y_i - \pi_i) = 0.$$

The equation for the intercept says that the residuals sum to zero: fitted probabilities total the observed event count. Aggregate calibration is therefore built into a converged, unpenalized logistic fit with an intercept; it cannot by itself establish a good model. The remaining equations need numerical solution because the probabilities depend nonlinearly on all coefficients.

```
m2 <- glm(event ~ sex + neuroticism + extraversion,
  data = dat, family = binomial(link = "logit"))
logLik(m2)
sum(fitted(m2))
```

The adjusted volunteering fit has log-likelihood -953.0306 . Directly summing its Bernoulli contributions gives the same value, and its fitted probabilities sum to 597, the observed number of volunteers. Appendix A connects this numerical fitting to scalar boundary cases, multiple starting values and local curvature.

3.2 Curvature, uncertainty and convergence

`glm()` uses **iteratively reweighted least squares** (IRLS): it forms a working linear regression with adjusted outcomes and weights $\hat{\pi}_i(1 - \hat{\pi}_i)$, updates the coefficients, and repeats. For a full-rank design without separation the logistic log-likelihood has a unique finite maximum. A stopping flag says the algorithm met its numerical criterion; the data and model still require scrutiny. This fit reports convergence after four iterations.

Before reading a covariance matrix, consider what the curve is measuring. In an invented two-group study, ten observations have $x = -1$ and ten have $x = 1$. Use the one-coefficient model $\text{logit}(p) = bx$, whose intercept is fixed at zero. If the groups contain 2 and 8 events, the likelihood peaks at $b = \log 4 = 1.3863$: moving either way makes the observed pattern less probable. If instead they contain 0 and 10 events, increasing b keeps improving the likelihood. Every finite curve leaves some chance of a contrary outcome, so the maximum is approached only as b diverges. This second pattern is **complete separation**.

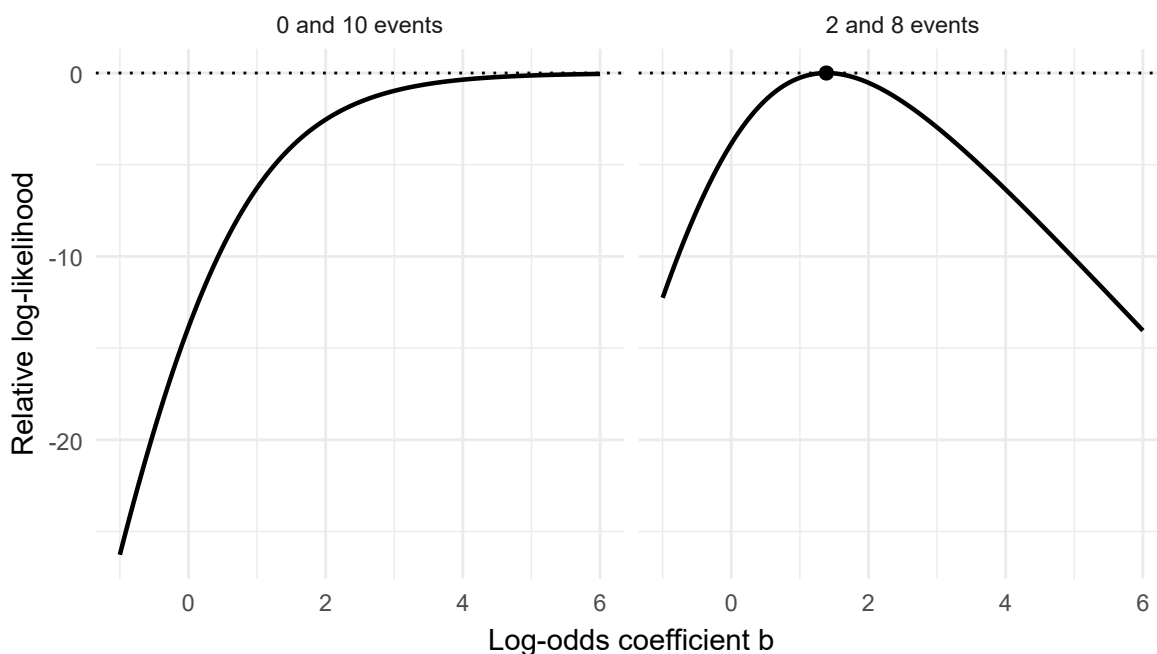


Figure 2: Changing the event counts changes whether the likelihood has a finite maximum.

The dashed zero line represents the best attainable log-likelihood in the finite case and the limiting value in the separated case. A flat tail is not evidence for a small coefficient; here it shows why a finite maximum is absent. Section 6.3 returns to this issue in the volunteering records.

Near the maximum, curvature describes how quickly fit deteriorates as a coefficient moves. Flat curvature allows many neighboring values and therefore larger standard errors. With design matrix $\mathbf{X}$ and diagonal weight matrix $\mathbf{W}$, the model-based covariance is

$$\widehat{\mathbf{V}} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}, \quad \mathbf{W} = \text{diag}\{\hat{\pi}_i(1 - \hat{\pi}_i)\}, \quad SE(\hat{\beta}_j) = \sqrt{\widehat{\mathbf{V}}_{jj}}.$$

A weight is largest at .5 and small near either boundary. Precision also depends on predictor spread and overlap, so neither total sample size nor event count alone determines it. Independent-row standard errors cannot repair dependence or measurement error.

A **Wald interval** is $\hat{\beta}_j \pm 1.96SE$ for 95% coverage under the large-sample Normal approximation, then exponentiated if the reported quantity is an odds ratio. A **profile-likelihood interval** instead keeps values not rejected by an appropriate likelihood-ratio comparison while optimizing the other coefficients. It can reflect asymmetry the Wald interval misses. The two approaches here differ by at most .0015 on the odds-ratio scale; they can diverge markedly in small samples or near separation (Hosmer et al., 2013).

```
m2$converged
m2$iter
exp(confint.default(m2))
exp(confint(m2))
```

A declared linear combination $C = \mathbf{c}^T \boldsymbol{\beta}$ has estimated variance $\mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}$. The off-diagonal covariance terms matter when combining coefficients, including for a predicted logit or a conditional slope. The familiar contrast rule remains; its covariance now comes from likelihood curvature.

3.3 A known process: certification success

Consider an invented certification course. Passing is the event; study hours span 0 to 20, and tutoring is an indicator with probability .35, independent of hours in this constructed process. Declare

$$\text{logit}\{P(\text{pass})\} = -2.2 + .3 \text{ hours} + .6 \text{ tutor}.$$

At seed 1010, 1,500 students give 961 passes. Fitting the declared model returns -2.2271 , $.3024$ and $.5264$, with all three generating values inside their Wald intervals. This is recovery under a known process, not evidence that an empirical study follows that process.

```
set.seed(1010)
hours <- round(runif(1500, 0, 20), 1)
tutor <- rbinom(1500, 1, .35)
pass <- rbinom(1500, 1, plogis(-2.2 + .30 * hours + .60 * tutor))
glm(pass ~ hours + tutor, family = binomial)
```

Table 6

One constant odds ratio, four different probability differences, in the known generating process.

Study hours	Probability	Probability at one more hour	Probability difference	Odds ratio
2	.1680	.2142	.0462	1.3499
7	.4750	.5498	.0748	1.3499
12	.8022	.8455	.0434	1.3499
17	.9478	.9608	.0130	1.3499

For students without tutoring, the declared one-hour odds ratio is $e^3 = 1.3499$ everywhere, yet the probability difference is $.0748$ at seven hours and $.013$ at seventeen. The starting probability makes the practical benefit different even before sampling variation enters. The estimated hours odds ratio, 1.3531 , closely recovers that truth.

Repeating the declared generation at seed 1012 for 500 samples of 300 gives an hours coefficient averaging $.3068$ with SD $.0359$ and empirical Wald coverage $.948$. The Monte Carlo SE of that coverage is $.0099$, so this run is compatible with $.95$ coverage. Finite-sample logistic estimates can be biased away from zero; simulation separates that tendency from variation in one sample. Tutoring's coefficient varies much more here (SD $.3309$): the two predictors have different scales and information, so their coefficient SDs should not be read as a general comparison of binary and continuous predictors.

The same generated data also let us compare mean functions honestly. The fitted linear probability model gives $.0492$ per hour but produces 193 fitted values above 1. Its fitted range is $.1253$ to 1.1861 .

This example deliberately uses a logistic truth and approaches the boundaries. It shows why an unbounded line can fail; it does not show that logistic regression must be right for every dataset. The volunteering analysis below supplies the complementary case in which both mean functions are numerically close over the observed range.

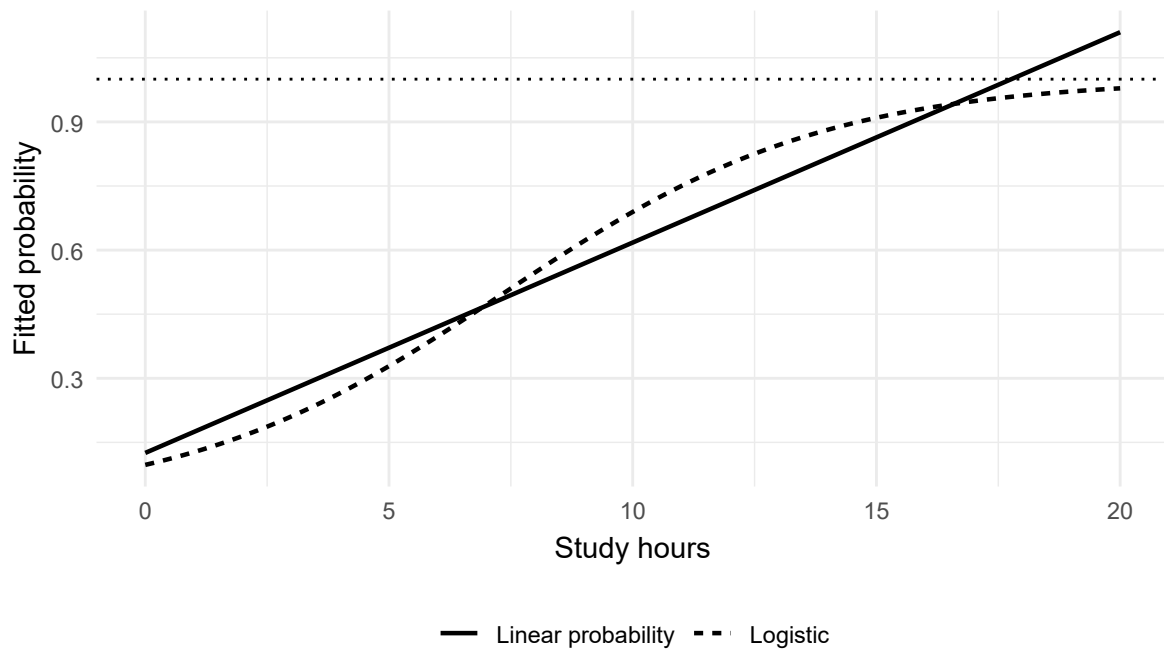


Figure 3: Two mean functions fitted to the same invented certification records, for students without tutoring.

4 Read Coefficients on Interpretable Scales

4.1 What an odds ratio compares

If a predictor enters additively, increasing it by one unit while holding the others fixed changes the log-odds by β_j :

$$\text{logit}\{\pi(x_j + 1)\} - \text{logit}\{\pi(x_j)\} = \beta_j, \quad \frac{\pi(x_j + 1)/\{1 - \pi(x_j + 1)\}}{\pi(x_j)/\{1 - \pi(x_j)\}} = e^{\beta_j}.$$

First fit extraversion alone. The coefficient is .0656 (SE .0141), giving an odds ratio of 1.0678 and a 95% interval [1.0386, 1.0978]. Each extra extraversion point is associated with 1.0678 times the odds of volunteering rather than not volunteering. Five points compound multiplicatively to 1.3883, rather than adding five copies of the odds ratio.

```
m1 <- glm(event ~ extraversion, data = dat, family = binomial)
exp(coef(m1))
exp(confint.default(m1))
```

Four qualifications make this a complete comparison: the predictor's **units**, any factor's **reference category**, the **event**, and the **adjustment set**. Changing a unit changes the contrast; changing the event or reference changes its direction; adding controls changes the conditional comparison. The intercept -1.1394 describes extraversion zero, outside the observed 2–23 range, so it is a mathematical anchor rather than a supported student profile.

An odds ratio is not a risk ratio. At a prevalence near .42, saying “1.5 times the odds” as “50 percent more likely” can substantially exaggerate a probability comparison. The approximation is useful only when the event probabilities in both groups are small.

The local slope on the probability scale is

$$\frac{\partial \pi}{\partial x_j} = \beta_j \pi (1 - \pi).$$

Its absolute value is bounded by $|\beta_j|/4$, attained at .5. This also bounds the magnitude of a one-unit finite probability difference when the logit is additive in that predictor. The derivative is a local rate, not exactly the difference between two profiles one full unit apart; compute both fitted probabilities when that finite contrast is wanted.

4.2 Reverse the event, preserve the underlying probabilities

Before looking at new output, predict the result of coding non-volunteering as 1. The probability must become $1 - \hat{\pi}_i$, the log-odds must change sign, and the odds ratio must become its reciprocal. The simple extraversion coefficient becomes $-.0656$ and its odds ratio .9365. Their signed z statistics reverse too. Standard errors, two-sided p-values, deviance and McFadden's pseudo R^2 stay the same. After translating back to the original event, every student still has the same fitted probability of volunteering.

```
dat$no <- 1 - dat$event
rev <- glm(no ~ extraversion, data = dat, family = binomial)
exp(coef(rev))
max(abs(fitted(rev) + fitted(m1) - 1))
```

Changing a predictor factor's reference instead changes coefficient coordinates while retaining the same outcome event and fitted probabilities. For a binary factor the focal comparison reverses, but its intercept moves and other coefficient changes depend on the model's terms. These recodings change what a printed coefficient names, not the fitted probability surface.

4.3 Adjusted results and probabilities at supported profiles

The multiple model is

$$\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 \text{male}_i + \beta_2 N_i + \beta_3 E_i,$$

where N and E are the recorded neuroticism and extraversion scores. Female is the reference sex category. The fitted table attaches each coefficient to that event, coding and adjustment set.

Table 7

Multiple logistic regression of volunteering on sex, neuroticism, and extraversion; Wald 95% intervals, female as the reference category, event coded as volunteering.

Term	Estimate (log-odds)	SE	z	Odds ratio	OR 95% lower	OR 95% upper
Intercept	-1.1165	.2491	-4.4829	.3274	.2010	.5335
Sex: male versus female	-.2352	.1112	-2.1151	.7904	.6357	.9829
Neuroticism	.0064	.0114	.5602	1.0064	.9842	1.0290
Extraversion	.0663	.0143	4.6512	1.0686	1.0391	1.0989

Among students of the same recorded sex and neuroticism, an extra extraversion point is associated with an odds ratio of 1.0686, close to the unadjusted 1.0678. Male students have .7904 times the odds of female students of the same personality scores. Neuroticism has no detected adjusted association in this additive model ($p = .5754$), but its interval admits odds ratios from .9842 to 1.0290 per point; a large p-value does not prove no relationship.

Choose profiles before interpreting their estimates: a female student at neuroticism 11, its median, and at extraversion 10, 13 or 15, its quartiles. These values sit within the observed support. Prediction first gives the logit and its SE; transform the two confidence limits to probability.

```
grid <- data.frame(sex = "female", neuroticism = 11,
  extraversion = c(10, 13, 15))
pred <- predict(m2, grid, type = "link", se.fit = TRUE)
plogis(cbind(pred$fit - 1.96 * pred$se.fit,
  pred$fit, pred$fit + 1.96 * pred$se.fit))
```

Table 8

Fitted probability of volunteering for a female student at the median neuroticism score, at the quartiles of extraversion, with intervals built on the log-odds scale and transformed.

Extraversion	Fitted log-odds	SE (log-odds)	Fitted probability	95% lower	95% upper
10	−.3833	.0830	.4053	.3668	.4451
13	−.1843	.0742	.4541	.4183	.4903
15	−.0516	.0813	.4871	.4474	.5269

The probability rises from .4053 to .4871 across the middle half of extraversion, a difference of .0818. Its interval is asymmetric because it is transformed from the logit scale. This is uncertainty about a conditional probability, not a prediction interval for an individual student: the next student’s outcome remains either 0 or 1. Comparing two separate intervals is also not a test of their difference; that contrast uses their joint covariance.

4.4 Profile and average probability comparisons

A **case-specific comparison** fixes the other predictor values and compares fitted probabilities at two profiles. An **averaged comparison** computes the contrast for each recorded student’s other predictor values and then averages it. This makes the averaging population explicit rather than treating one artificial “average student” as representative.

Averaging is a substantive choice even when the model is already fixed. Consider the invented model $p(x, u) = \text{logit}^{-1}(x + u)$ and compare $u = 1$ with $u = 0$. One population has equal numbers at $x = -.5$ and $.5$; another has equal numbers at $x = -4$ and 4 . Both have mean zero. Predict which population has the larger average probability difference from the vertical gaps in Figure 4.

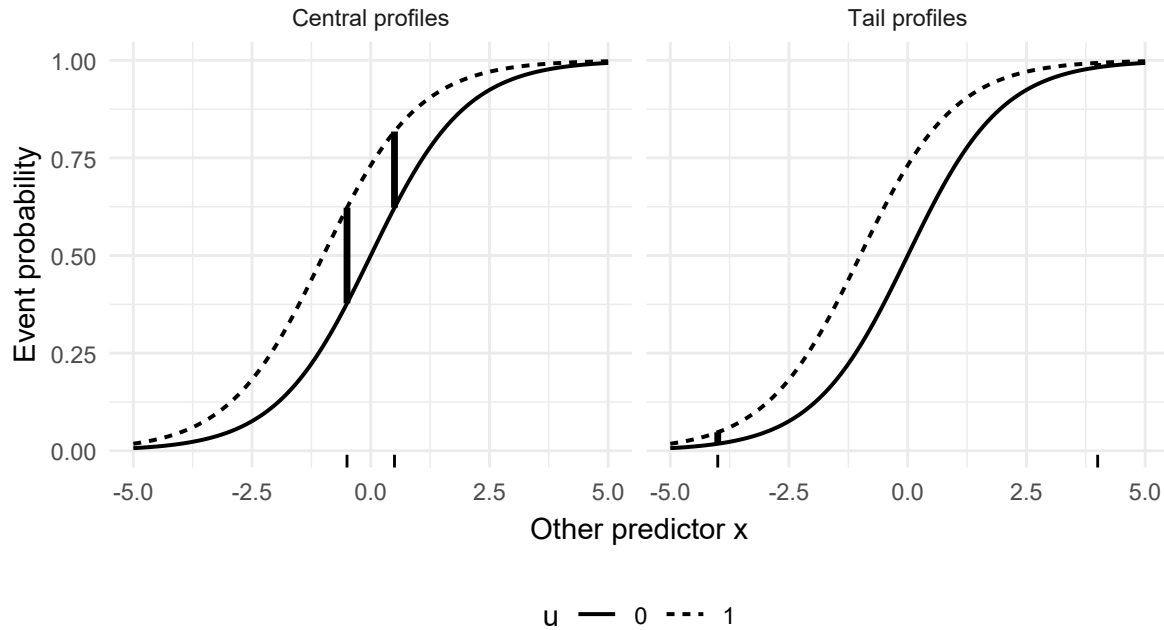


Figure 4: The same probability model gives different average differences in two populations.

The central profiles give average difference .22 and the tail profiles give .0204. In contrast, evaluating both at the common mean $x = 0$ gives .2311. The coefficient of u and its conditional odds ratio e are identical throughout. The average changes because the populations place different weight on steep and flat portions of the same probability curves. A profile at the average predictor is therefore not interchangeable with an average over the actual profiles. These are model comparisons, not causal effects supplied by the invented equation.

```
x <- c(-.5, .5)
mean(plogis(x + 1) - plogis(x))
x <- c(-4, 4)
mean(plogis(x + 1) - plogis(x))
plogis(1) - plogis(0)
```

Return now to the recorded volunteering profiles.

```
plus <- dat
plus$extraversion <- plus$extraversion + 1
mean(predict(m2, plus, type = "response") - fitted(m2))
```

The average one-point extraversion difference is .0159. The local maximum bound is .0166, and an additive linear probability fit gives .0158. Here its fitted values remain within [.2261, .6188], near the middle of the logistic curve, so the two mean functions agree closely. The familiar least-squares constant-variance uncertainty still needs correction, and a new predictor range can expose the line's boundaries. The one-point standardization also moves the single student at score 23 to 24, just outside the recorded range; that small extrapolation is part of this explicitly defined average, not evidence about students at 24.

Assigning each observed profile first the female and then the male indicator gives an averaged male-minus-female probability difference of $-.0562$. This is about 5.6 percentage points, whereas the odds ratio .7904 describes odds about 21 percent lower. The distinction changes what a reader imagines about actual volunteering rates.

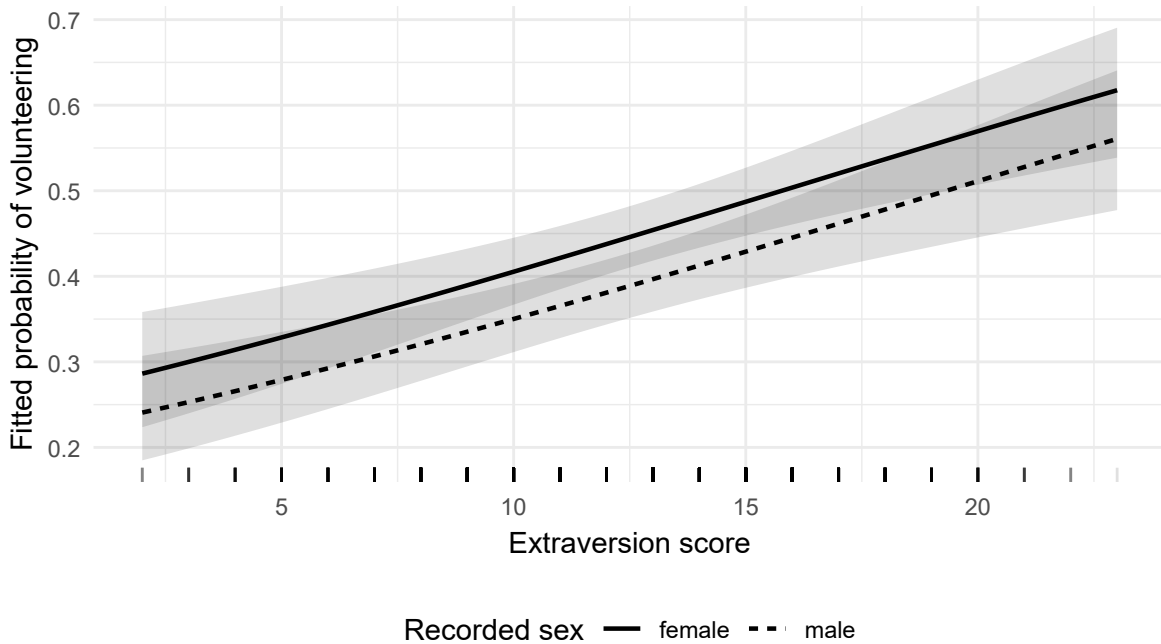


Figure 5: Adjusted volunteering probabilities at neuroticism 11, with pointwise Wald 95% bands and observed score support.

The two lines are parallel on the logit scale because the additive model contains no product term. Their probability separation varies. Additivity belongs to a named scale, a distinction that matters when we next allow conditional contrasts to differ.

5 Compare and Test

5.1 Deviance and a declared model comparison

The **deviance** compares a fitted model with the saturated model that fits every observation exactly. For individual binary observations the saturated log-likelihood is zero, so

$$D = -2\ell(\hat{\beta}), \quad G^2 = D_A - D_B \sim \chi_q^2 \quad \text{for regular nested models } A \subset B.$$

Here q counts the restrictions defining the smaller model. The comparison requires the same rows and event, finite well-identified estimates, and an adequate large-sample approximation. The absolute ungrouped binary deviance has no simple chi-square goodness-of-fit reading; its difference can test a declared nested comparison.

```
anova(m0, m2, test = "LRT")
small <- glm(event ~ sex + extraversion, data = dat, family = binomial)
anova(small, m2, test = "LRT")
```

Table 9

Two declared likelihood-ratio comparisons on the same 1,421 rows and the same event.

Comparison	Added	df	G^2	p
Intercept only versus full	Sex, neuroticism, extraversion	3	27.4447	< .0001
Without neuroticism versus full	Neuroticism	1	.3139	.5753

The full block reduces deviance from 1933.5060 to 1906.0613. Its statistic 27.4447 exceeds the 5% critical value 7.8147 on three df. This establishes information in the block under the model, not a separate association for each member. Removing neuroticism alone costs only .3139 on one df, with $p = .5753$.

5.2 Wald tests and likelihood comparisons

A single-coefficient Wald test uses $z = \hat{\beta}/SE$, or z^2 with a chi-square reference on one df. The likelihood-ratio test instead compares the best fits allowed by two models. For neuroticism the two agree: $z = .5602$, $z^2 = .3138$, $p = .5754$, against $G^2 = .3139$, $p = .5753$. Close agreement is expected when the likelihood is nearly quadratic near its maximum. With strong asymmetry or unstable estimates, the Wald approximation can be misleading. Separation below shows an extreme case in which its statistic shrinks despite a perfect pattern.

5.3 Criteria rank models; they do not certify them

For k estimated coefficients and n records,

$$AIC = -2\ell + 2k, \quad BIC = -2\ell + k \log n, \quad R_{\text{McF}}^2 = 1 - \frac{\ell}{\ell_0}.$$

There is no separately estimated variance parameter in the logistic model. AIC and BIC penalize in-sample likelihood by model size; smaller values are preferred among models fitted to the same records and outcome (Akaike, 1974; Schwarz, 1978). McFadden's pseudo R^2 compares with the intercept-only maximum ℓ_0 . It is not a proportion of variance explained and should not be read against linear-model R^2 benchmarks.

```
AIC(m0, m1, m2)
BIC(m0, m1, m2)
1 - as.numeric(logLik(m2) / logLik(m0))
```

Table 10

Model comparison for the volunteering analysis; both criteria are comparative and neither certifies adequacy.

Model	k	Deviance	AIC	BIC	McFadden pseudo R^2
Intercept only	1	1933.5060	1935.5060	1940.7651	.0000
Extraversion	2	1911.4838	1915.4838	1926.0020	.0114
Sex, neuroticism, extraversion	4	1906.0613	1914.0613	1935.0977	.0142

AIC prefers the adjusted model; BIC prefers extraversion alone. At $n = 1421$, BIC charges $\log n = 7.2591$ per coefficient against AIC's 2, explaining the difference. Neither criterion is a test or a held-out prediction score. The small McFadden value .0142 is compatible with a detectable association and little ability to distinguish volunteers.

5.4 A product term changes a conditional contrast

Allow the extraversion contrast to vary with neuroticism:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{male}_i + \beta_2 N_i + \beta_3 E_i + \beta_4 N_i E_i, \quad \frac{\partial \text{logit}(\pi)}{\partial E} = \beta_3 + \beta_4 N.$$

The product coefficient is $-.0086$ (SE $.0029$). The nested likelihood comparison gives $G^2(1) = 8.6213$, $p = .0033$. The lower-order extraversion coefficient $.1668$ is its contrast at neuroticism zero, not a summary for every student. The additive model's undetected neuroticism term and the detected product answer different questions, so hierarchical model construction retains both lower-order terms.

```
m3 <- glm(event ~ sex + neuroticism * extraversion,
  data = dat, family = binomial)
anova(m2, m3, test = "LRT")
```

Table 11

Conditional extraversion odds ratios at the quartiles of neuroticism, each comparing students differing by one extraversion point at the stated neuroticism score.

Neuroti- cism	Log-odds contrast	SE	Odds ratio	OR 95% lower	OR 95% upper	Students within one point
8	.0984	.0184	1.1034	1.0644	1.1438	276
11	.0727	.0146	1.0755	1.0451	1.1067	302
15	.0385	.0170	1.0393	1.0052	1.0746	242

The table evaluates $\beta_3 + \beta_4 N$ at supported quartiles, with SEs from $\mathbf{c}^\top \widehat{\mathbf{V}} \mathbf{c}$. At extraversion 13, the probabilities for neuroticism 8 and 15 are nearly equal ($.4519$ and $.4512$), but a one-point extraversion difference is $.0245$ at the former and $.0096$ at the latter. Conversely, different probability differences can arise with a constant odds ratio simply because baseline probabilities differ. An interaction claim therefore names its scale. Chapter 10 develops centering and probing; the new issue here is the nonlinear transfer between scales.

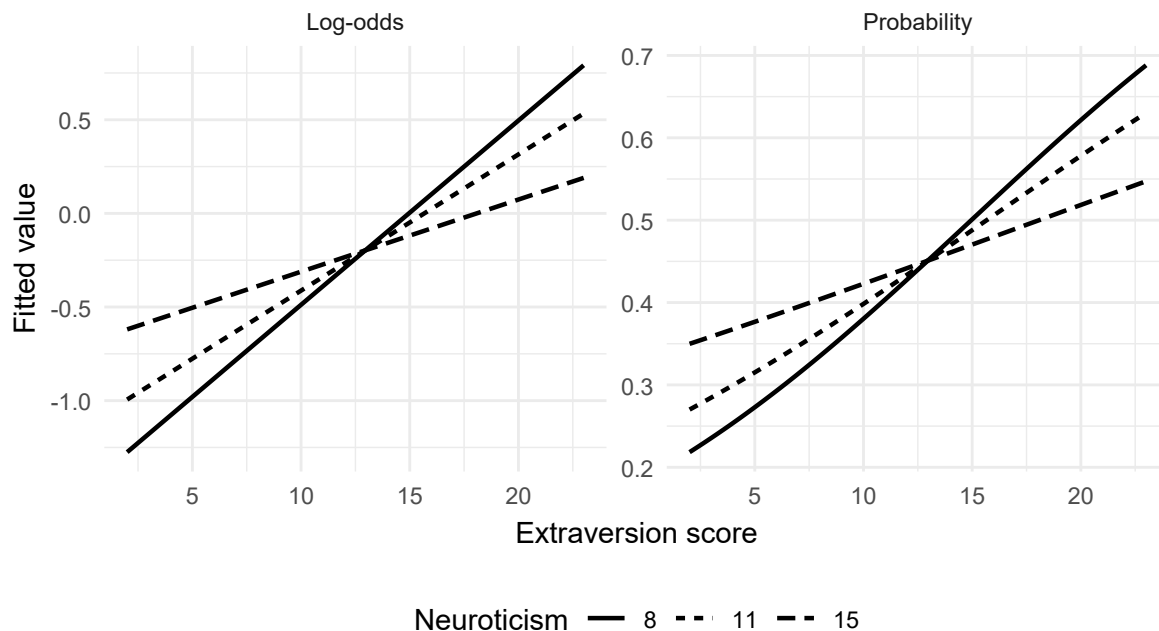


Figure 6: The same product-term fit on log-odds and probability scales, for female students.

6 Check Calibration, Fit, and Influence

6.1 Form and observed support

An observed proportion at each extraversion score makes both the pattern and its support visible. An **empirical logit** $\tilde{\eta} = \log\{(y + .5)/(n - y + .5)\}$ adds half a count to keep groups with zero or all events finite. It is a plotting device rather than a replacement for the fitted Bernoulli likelihood.

```
aggregate(event ~ extraversion, data = dat, mean)
anova(m2, mq, test = "LRT")
```

Score 13 has 151 students. Score 22 has five, all volunteers; score 23 has one non-volunteer. Their extreme empirical logits, 2.3979 and -1.0986 , deserve less weight than the densely populated middle. The observed proportions need not increase at every score: sampling variation around an increasing conditional mean remains.

The grouped pattern gives no compelling curvature, and adding squared extraversion gives $G^2(1) = 1.4257$, $p = .2325$. This does not prove linearity in the logit; it says this particular alternative is not clearly distinguished by the available records. A one-predictor empirical-logit

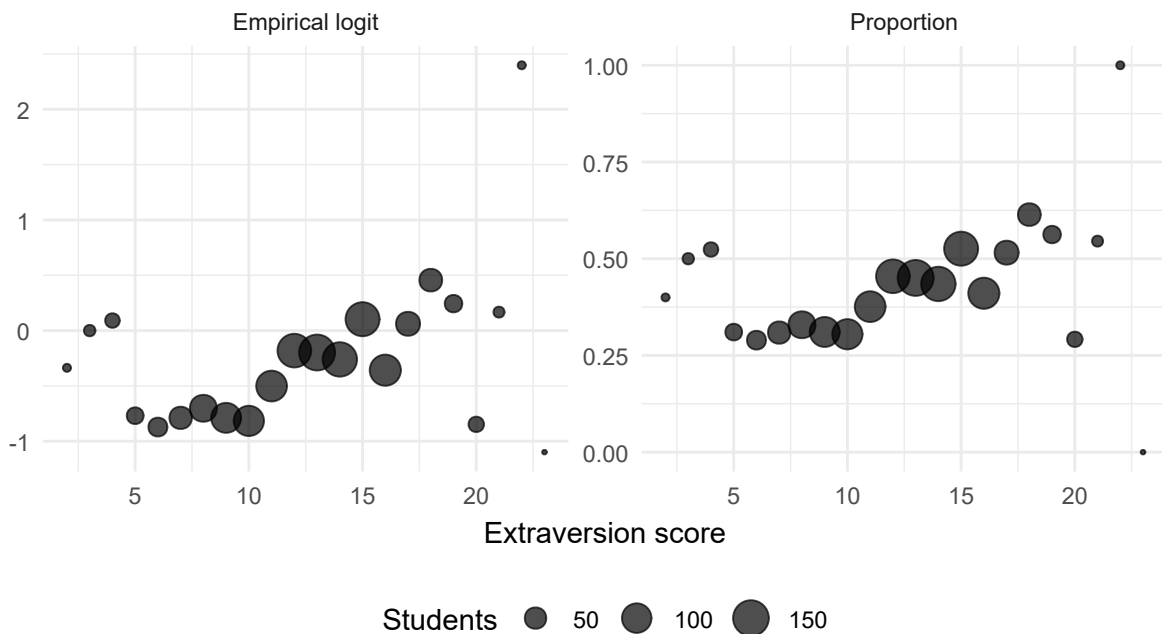


Figure 7: Observed volunteering by score: empirical proportions and logits, with point area showing the number of students.

display is also a marginal diagnostic, not a complete check of the adjusted model's functional form.

6.2 Residuals and influence

Raw residuals are $y_i - \hat{\pi}_i$. At a fitted probability they take only $1 - \hat{\pi}_i$ or $-\hat{\pi}_i$, so the raw plot has two lines rather than a Normal cloud. Pearson residuals divide by $\sqrt{\hat{\pi}_i(1 - \hat{\pi}_i)}$; signed deviance residuals have squares summing to deviance. None is expected to be Normal.

A **binned residual plot** instead groups nearby fitted probabilities and compares their average residuals with zero. The displayed pointwise bands, $\pm 2\sqrt{\bar{\pi}(1 - \bar{\pi})/n_{\text{bin}}}$, are a rough visual reference for groups with similar fitted probabilities; they are not a simultaneous test and ignore parameter-estimation effects.

One bin falls outside its reference band. That is compatible with chance variation and does not certify the form. The largest Cook's distance is .0051. Removing that row moves the extraversion coefficient from .0663 to .0679, with a maximum coefficient change of .0304 elsewhere. A deletion refit asks whether a conclusion depends on one observation; it supplies no automatic reason to discard it. The principal analysis retains the row.

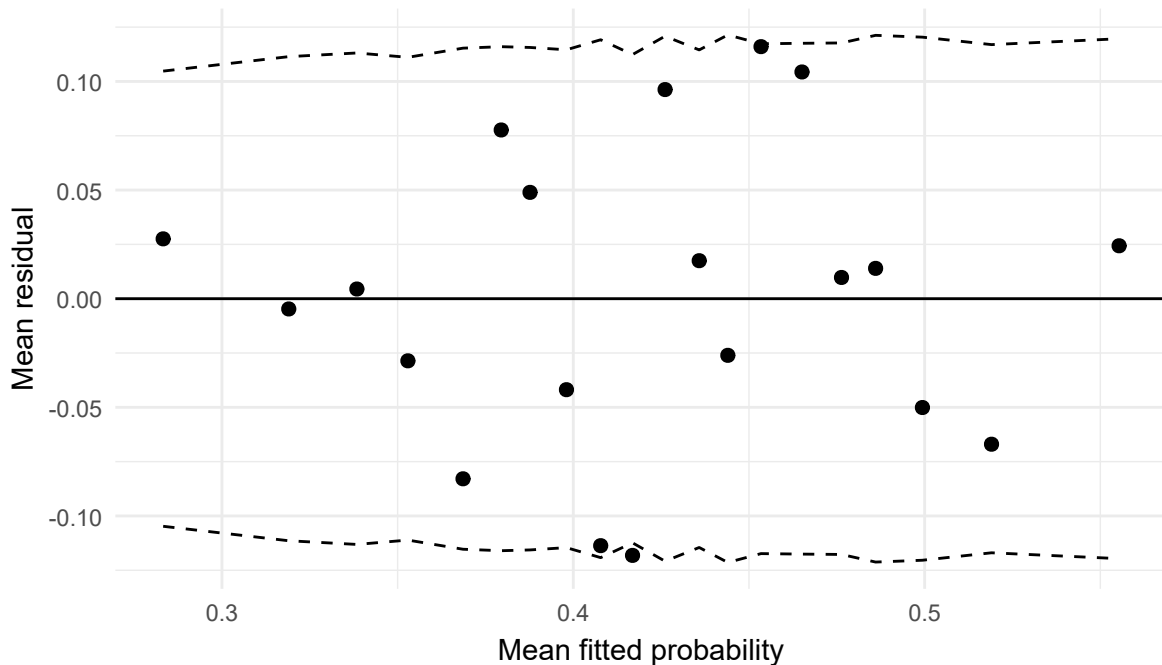


Figure 8: Mean residuals in twenty fitted-probability bins, with approximate pointwise two-SE reference bands.

```
which.max(cooks.distance(m2))
coef(m2)
coef(md)
```

6.3 Separation: a strong pattern with no finite estimate

A sparse category supplies little information. **Complete separation** is more specific: a predictor combination places every event on one side and every non-event on the other. With **quasi-complete separation**, some observations lie on the separating boundary. In these circumstances one or more coefficients can increase in magnitude without reaching a finite likelihood maximum (Albert & Anderson, 1984). A number printed by software can be merely the point at which iteration stopped.

Make this failure visible with a deliberately constructed subset, not a substantive analysis. Retain non-volunteers scoring at most 8 and volunteers scoring at least 17, then sample 60 eligible rows at seed 1011. There is a gap between the two outcomes by construction.

```
glm(event ~ extraversion, data = sep, family = binomial)
```

Table 12

Symptoms of this complete-separation example, read together.

Output feature	Constructed separated fit	What it signals here
Extraversion coefficient	5.5928	An iteration moving toward an infinite limit
Standard error	8014.9006	Nearly flat local information in the escaping direction
Wald z and two-sided p	.0007; .9994	A failed Wald approximation, not absence of the observed pattern
Residual deviance	5.642×10^{-10}	Almost perfect fit on the selected rows
Fitted probabilities	about 2.22×10^{-16} and 1	Numerical boundary values
Iterations	25	Far more than the four needed by the regular adjusted fit

The selected rows reproduce the separated pattern in Figure 2: increasing the slope can approach perfect classification without a finite likelihood peak.

The tiny Wald statistic is particularly instructive: it does not negate the perfect ordering. The coefficient and SE have both become enormous, and the usual local approximation is unusable. These are characteristic symptoms in this example, not a checklist every separated model must satisfy. Quasi-complete separation need not make the whole deviance zero, and a warning is neither necessary nor sufficient for diagnosis.

The full data offer an unwarned example. If scores 22 and 23 are separate factor levels, one contains five events and the other one non-event. Their coefficients become 15.2161 (SE 394.7749) and -13.916 (SE 882.7434). Combining the substantively adjacent upper bands as 18+ gives 130 students with 71 events and a finite coefficient .8352 (SE .2222). That demonstrates what the sparse bands were doing; such a merge still requires a meaningful category definition, not merely a wish for smaller standard errors.

Table 13

Options when a logistic model shows separation.

Option	What it does	When it is appropriate
Collapse categories	Merges a sparse level with a substantively adjacent one	The merged levels are defensibly similar on subject-matter grounds

Option	What it does	When it is appropriate
Drop the term	Removes the separating predictor	The predictor is not central to the question and its removal is reported
Report the boundary honestly	States the direction, notes that the magnitude is not identified, and reports any unbounded profile limit	The separating pattern is itself the finding
Penalized likelihood	Adds a penalty giving finite estimates with reduced small-sample bias, as in Firth's method	A finite estimate is needed and the penalty is disclosed
Collect more data	Fills the empty cell	The design permits it

Firth's penalized likelihood is one way to obtain finite estimates with reduced small-sample bias (Firth, 1993). It is a different estimation rule, whose penalty must be stated. It does not make the original infinite maximum-likelihood estimate finite. More generally, a finite estimate should be interpreted only where the data support the intended comparison.

6.4 Calibration and discrimination

Calibration asks whether assigned probabilities agree with observed event frequencies: among students assigned about .4, do roughly 40% volunteer? **Discrimination** asks whether probabilities tend to rank volunteers above non-volunteers. They answer distinct questions. A constant fitted prevalence has perfect aggregate agreement but ranks nobody; a rule can rank well and systematically overstate probabilities.

The fifth decile is assigned .4126 but has observed proportion .2966, a gap of .1161 or 3.0596 observed-proportion SEs. This deserves investigation rather than dismissal. Because the group was noticed among ten and the model was fitted to these same rows, this descriptive SE comparison is not a calibrated global test of fit. Aggregate agreement from the intercept says nothing about agreement within each decile.

Fitted probabilities average .4313 among volunteers and .412 among the rest, a difference of .0192, and span only [.2418, .6235]. The boxes overlap strongly. Extraversion is detectably associated with volunteering, yet the fitted probabilities separate these students poorly. A small coefficient p-value is not evidence of useful prediction (Shmueli, 2010). Both displays reuse the fitting data and are optimistic as performance assessments. Chapter 13 develops held-out evaluation, including AUC, Brier score, log loss and threshold choice.

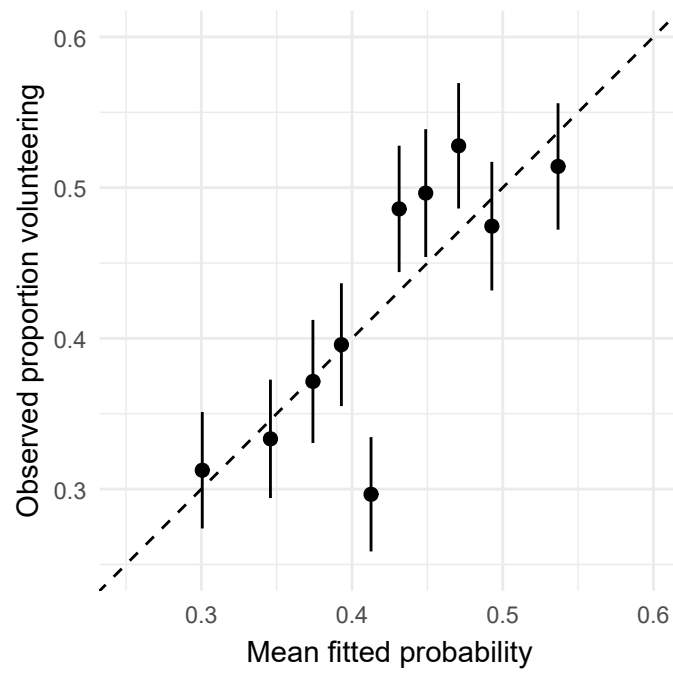


Figure 9: In-sample calibration by fitted-probability decile; vertical segments show one observed-proportion SE.

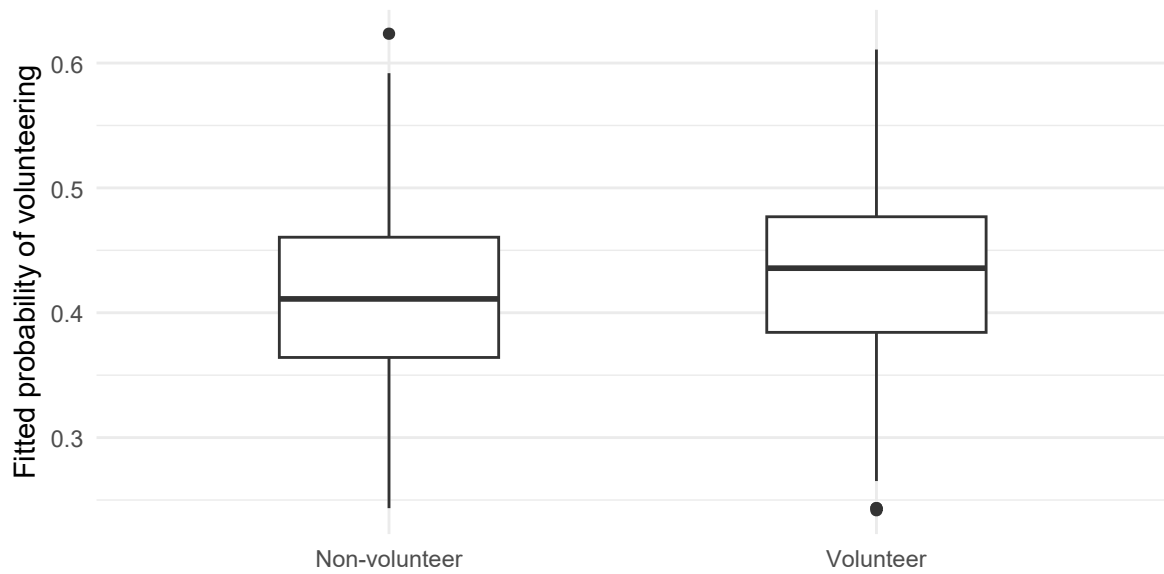


Figure 10: Fitted probabilities overlap strongly between volunteers and non-volunteers.

6.5 What changed under sensitivity analysis

Table 14

Sensitivity of the extraversion odds ratio to modeling and sample choices; each row refits the model with one feature changed.

Specification	Extraversion odds ratio per point
Simple model, extraversion only	1.0678
Additive model, sex and neuroticism added	1.0686
Additive model, most influential row removed	1.0703
Additive model plus quadratic extraversion, local at extraversion 13	1.0745
Product-term model, at median neuroticism	1.0755

The quadratic entry is a local derivative-based odds ratio at score 13; it is not exactly the finite 13-to-14 contrast. The product entry fixes neuroticism at its median. These definitions keep the compared quantities clear. The values remain close, but every specification uses the same underlying study, measurements and design. Stability across them cannot establish accurate volunteering records, valid personality measurement, independent classrooms or absence of unmeasured common causes.

7 From Analysis to a Defensible Claim

7.1 Report the magnitude and the limit together

The event and analytic unit come before any ratio. The model, predictor units and reference categories make the comparison reproducible. A probability-scale result then says what the ratio means in this study, while checks and uncertainty qualify the conclusion.

A bounded reporting example

Among 1,421 recorded introductory psychology students, the volunteering proportion was .4201. Adjusting for recorded sex and neuroticism, each extraversion point was associated with 1.0686 times the odds of volunteering rather than not volunteering (95% CI [1.0391, 1.0989]). The averaged one-point probability difference was .0159 over the observed profiles. For a female student at neuroticism 11, fitted probabilities were .4053 [.3668, .4451] at extraversion 10 and .4871 [.4474, .5269] at 15. The additive neuroticism association was not detected, although a neuroticism-by-extraversion product improved fit and showed a weaker extraversion contrast at higher neuroticism. A quadratic check and a

single-row deletion gave similar local comparisons. Sparse extreme scores limited their own interpretation; one calibration decile showed a material departure, and fitted probabilities overlapped strongly by outcome. These are conditional associations among recorded students. They establish neither a causal effect of personality nor useful prediction in a new cohort.

The report can contain a null comparison, a detected interaction and weak discrimination together. They address different aspects of the research question. Describing them faithfully is more useful than converting a coefficient table into a general claim that the model “worked”.

7.2 Recognize the claim a number supports

Table 15

Interpretation depends on the quantity and its conditions.

Tempting sentence	What the evidence supports
“Volunteering is about 7% more likely per extraversion point.”	Odds are 1.0686 times as large; the averaged probability difference is .0159.
“Neuroticism has no relationship with volunteering.”	The additive adjusted interval admits small associations in both directions; the product term asks another question.
“The product coefficient is the extraversion effect.”	Evaluate the conditional contrast at a stated neuroticism value and name its scale.
“The small p-value means the model predicts well.”	A coefficient test, calibration check and held-out performance assessment answer different questions.
“The separated fit found a very large, imprecise effect.”	Ordinary maximum likelihood has no finite estimate for that separated magnitude.

A reusable report makes seven elements recoverable: event and unit; model and coding; a coefficient-scale estimate with uncertainty; a probability-scale comparison; evidence for a declared term or block; adequacy and sensitivity; and the claim’s boundary. They need not appear as seven ritual sentences. Their function is to let a reader judge what was estimated and why its interpretation is warranted.

The empirical analyses use all Cowles records, except the explicitly labelled deletion and constructed separation checks. The certification sample uses seed 1010, its repetition uses 1012, and the separated subset uses 1011. More than two categories, event counts, dependence, severe measurement error and sparse events each require a reconsidered model or inferential calculation, not a stronger reading of this coefficient table.

8 Reading Bridge

A logistic model connects a Bernoulli event to a linear predictor through the logit link. Maximum likelihood supplies estimates and uncertainty; probabilities give the substantive magnitude. Event coding, support, separation and the distinction between calibration and discrimination remain part of interpreting every result. Chapter 12 extends the same framework to nominal categories, ordered categories and counts, each with a new response structure and a new meaning for an exponentiated coefficient.

Key terms

This reference collects the concepts introduced here. The Part One Concept Reference locates the broader ideas, including likelihood, coefficient covariance and model comparison.

Table 16

Concepts introduced in this chapter.

Term and common alternatives	Meaning or essential distinction
Event and non-event (the category coded 1)	The category a binary model treats as occurring, named together with the category coded 0. Reversing the event reciprocates the odds ratios and complements fitted event probabilities; probabilities for the original event agree after relabelling.
Prevalence (event rate)	The proportion of analytic rows that are events. Coefficient precision also depends on conditional probabilities, predictor spread and overlap; the rare-outcome approximation of an odds ratio by a risk ratio depends on the risks being compared.
Generalized linear model (GLM)	A response distribution, a linear predictor and a link connecting the conditional mean to the linear predictor; changing the response distribution and link gives the models of Chapters 11 and 12.
Link function; logit link and inverse logit (logistic function)	The logit is the log of the odds and its inverse the S-shaped function bounded in $(0, 1)$, equal to one half and steepest at a linear predictor of zero.
Odds and log-odds	The odds are $\pi/(1 - \pi)$ and the log-odds their logarithm; the three scales describe one quantity with null values 1 and 0, and the log-odds scale is symmetric under reversing the event.

Term and common alternatives	Meaning or essential distinction
Odds ratio (OR)	The factor e^{β_j} by which the odds differ between units differing by one unit in a predictor and agreeing on the others; it depends on the units, reference, event and adjustment set, is not a risk ratio unless the event is rare, and implies no constant probability difference.
Noncollapsibility (of the odds ratio)	A conditional odds ratio generally differs from the marginal one even with no confounding, so a coefficient that moves under adjustment is not by itself evidence of confounding.
Bernoulli likelihood and score equations (binary-outcome likelihood)	The probability of the observed data as a function of the parameters, maximized to estimate them; for a binary outcome it is a product of Bernoulli probabilities whose score equations have no closed form, and a model with an intercept reproduces the observed event total.
Iteratively reweighted least squares (IRLS)	The algorithm that repeatedly solves a weighted linear regression with weights $\hat{\pi}_i(1 - \hat{\pi}_i)$ until the coefficients stop changing; its convergence flag and iteration count belong in the record and establish no adequacy.
Deviance and the likelihood-ratio test (G^2)	For ungrouped binary data the deviance is -2 times the maximized log-likelihood and carries no goodness-of-fit reading alone, while the difference between nested models on the same rows and event is referred to chi-square.
Wald test and profile-likelihood interval	The Wald statistic squares the estimate-to-SE ratio while a profile interval collects the values the log-likelihood does not reject; they diverge for rare events, small samples or near-infinite estimates.
McFadden pseudo R^2	One minus the ratio of the fitted to the intercept-only maximized log-likelihood: bounded in $[0, 1]$, not a proportion of variance explained, and routinely small for useful models.
Empirical logit	The log-odds of a grouped event count with one half added to each count, plotted against a predictor to check linearity of the log-odds in that predictor as entered.
Binned residual	The mean raw residual within a bin of rows sorted by fitted probability, against a two-standard-error band; it replaces the residual-versus-fitted plot, which shows only two curved bands here.
Separation (complete; quasi-complete)	A predictor pattern separates events from non-events completely or on a boundary, so at least one coefficient has no finite ordinary maximum-likelihood estimate. Diverging coefficients, large standard errors and convergence warnings can be symptoms; no one warning or numerical pattern is required.

Term and common alternatives	Meaning or essential distinction
In-sample calibration and discrimination checks (fitted-probability checks)	Calibration asks whether fitted probabilities match observed event frequencies, discrimination whether they rank events above non-events; the two answer distinct questions, and honest evaluation of new-case performance requires held-out data.
Linear probability model (LPM)	Ordinary least squares on a 0/1 outcome, whose coefficients are probability differences but whose mean function is unbounded and whose conditional variance cannot be constant.
Averaged probability difference (average marginal comparison)	The mean, over the analytic rows with their own other predictor values, of the difference between fitted probabilities at two values of one predictor.

Further reading

Agresti (Agresti, 2018) and Hosmer, Lemeshow and Sturdivant (Hosmer et al., 2013) develop logistic models and diagnostics. Gelman, Hill and Vehtari (Gelman et al., 2020) emphasize probability-scale interpretation. The R `glm()`, `family()` and `predict.glm()` documentation explains the implementation (R Core Team, n.d., 2025).

Exercises

Exercise 1: Read a logistic result on three scales (Foundational).

A logistic regression of whether a student volunteered for an experiment on an extraversion score in integer points, adjusting for recorded sex and a neuroticism score, gives an extraversion coefficient of .0663 with standard error .0143. The event is volunteering, the reference sex category is female, extraversion runs from 2 to 23, and the prevalence is .4201. (1) Give the odds ratio and its Wald 95% interval, and write one sentence interpreting them with the units, the event and the adjustment set explicit. (2) Give the odds ratio for a five-point difference. (3) Give the divide-by-four bound on the probability difference per point, compute the local rate of change at a fitted probability of .12 from $\partial\pi/\partial x = \beta\pi(1-\pi)$, and explain why it must be smaller than the bound. (4) Name three conclusions the interval excluding 1 does not establish.

Exercise 2: Declare the event, then reverse it (Foundational).

A researcher codes program *completion* as the event and reports an odds ratio of .62 per additional hour of paid weekly employment, with a 95% interval of [.51, .75]. (1) State in one sentence what this odds ratio compares, including the event and the units. (2) Give the odds ratio and interval the same fitted model would report with *non-completion* as the event, showing the arithmetic. (3) State which of these change under the recoding and which do not: the coefficient signs, the standard errors, the z statistics, the p -values, the deviance, the fitted probability of completion for a given student, and McFadden's pseudo R^2 . (4) The researcher also relevels the employment-status factor so that "employed" is the reference, and a reviewer complains that the results changed. Explain in what sense they did and did not change.

Exercise 3: Odds ratios are not probability differences (Foundational).

Two studies report the same adjusted odds ratio of 2. In Study A the comparison group's event probability is .04; in Study B it is .45. (1) For each study compute the exposed group's event probability, the probability difference and the risk ratio. (2) Explain why the odds ratio and the risk ratio nearly coincide in one study and diverge in the other. (3) A policy brief states that "an odds ratio of 2 means the risk is doubled" for both. Say where this is approximately acceptable and where it is not, and quantify the error. (4) Explain why noncollapsibility means an adjusted odds ratio can differ from an unadjusted one even when the added predictor is unrelated to the focal predictor. (5) Recommend, with reasons, which scale you would lead with in a report for school administrators, and which you would also supply. (6) In the invented model $p(x, u) = \text{logit}^{-1}(x + u)$, compare $u = 1$ with $u = 0$. Population C has equal numbers at $x = -.5$ and $.5$; Population D has equal numbers at $x = -4$ and 4 . Before calculating, predict which has the larger average probability difference. Explain why their common mean $x = 0$ and common conditional odds ratio do not make these averages equal.

Exercise 4: Deviance, Wald, and the criteria by hand (Foundational).

On 1,421 rows with the same event, an intercept-only logistic model has deviance 1933.506 and a model with sex, neuroticism and extraversion has deviance 1906.0613 with four fitted coefficients; the 5% critical value of χ^2_3 is 7.8147 and $\log 1421 = 7.259116$. The neuroticism coefficient is .0064 with standard error .0114. (1) Calculate the likelihood-ratio statistic for the three-predictor block, give its df, and state the decision at 5% and what it does and does not say about the three added columns. (2) Calculate AIC and BIC for the fuller model and say which model each criterion prefers, given the single-predictor model's AIC 1915.4838 and BIC 1926.002. (3) Calculate the Wald statistic for neuroticism, compare it with the likelihood-ratio value .3139, and say when the two would be expected to disagree. (4) Explain why a deviance of 1906.0613 on its own says nothing about fit.

Exercise 5: Evaluate a product term, do not read it (Integrative).

A logistic model of volunteering contains sex, neuroticism N , extraversion E and their product, with $\hat{\beta}_E = .1668$ and $\hat{\beta}_{NE} = -.0086$; neuroticism runs from 0 to 24 with quartiles 8, 11 and 15, the likelihood-ratio test of the product gives $G^2(1) = 8.6213$, $p = .0033$, and neuroticism alone had $p = .5754$ in the additive model. (1) Give the conditional extraversion log-odds contrast and odds ratio at each quartile. (2) Explain what .1668 refers to and why it is not “the extraversion effect”. (3) Explain how a term with no detected additive association can participate in a detected product, and why the additive term is kept. (4) At extraversion 13 a female student’s fitted probability is .4519 at neuroticism 8 and .4512 at 15, and the one-point probability differences are .0245 and .0096. Explain why the probability and log-odds pictures can agree here and disagree elsewhere, and what a report must name before claiming an interaction.

Exercise 6: Read a separated output and a calibration display (Integrative).

A logistic regression fitted to 60 rows returns a coefficient of 5.5928 with standard error 8014.9006, $p = .9994$, a residual deviance of 5.642×10^{-10} , fitted probabilities of essentially 0 and 1, and 25 iterations against the four a well-behaved model needed. A separate, well-behaved model has one calibration decile with mean fitted probability .4126 against an observed proportion of .2966, whose standard error is .0379, while the other nine lie within 1.4 standard errors; its fitted probabilities span .2418 to .6235. (1) Name the pattern producing the first output and say why the Wald test reports no evidence for the strongest possible pattern. (2) State what a report can and cannot say about that coefficient, and name two remedies with the condition each requires. (3) Express the calibration gap in standard errors and say how much weight one such decile deserves among ten. (4) State what the fitted-probability range says about discrimination, why that differs from calibration, and why both are optimistic when computed on the fitting rows. (5) Sketch the qualitative likelihood shape for a finite estimate and for the separated example. Explain why a finite stopping point and a very small change in likelihood do not establish a finite maximum.

Appendix A: Likelihood Mathematics and Numerical Checks

A numerical optimum, a finite identified parameter and an adequate scientific model are different claims. The following two examples make their checks concrete.

Interior and boundary solutions

For s successes in $n > 0$ independent Bernoulli trials with common probability p ,

$$L(p) = \binom{n}{s} p^s (1-p)^{n-s}, \quad \ell(p) = \log \binom{n}{s} + s \log p + (n-s) \log(1-p).$$

For $0 < s < n$, differentiating on $0 < p < 1$ gives

$$\ell'(p) = \frac{s}{p} - \frac{n-s}{1-p}, \quad \hat{p} = \frac{s}{n}.$$

The binomial coefficient is constant in p , so recording only the count rather than the particular observed 0/1 pattern does not change the maximizing p : the Bernoulli product $p^s(1-p)^{n-s}$ and the binomial likelihood share a maximizer. With only an intercept in a logistic model, $p = \text{logit}^{-1}(\beta_0)$, so the interior solution is $\hat{p} = s/n$ with $\hat{\beta}_0 = \text{logit}(s/n)$. At $s = 0$ or $s = n$ the likelihood is maximized at a boundary probability of 0 or 1 and there is no finite intercept estimate, which is the one-parameter form of the separation problem Section 6.3 develops for predictor patterns; the analogy motivates, and does not replace, that section's conditions. Evaluate the boundaries explicitly, because an interior numerical search can return a point near an endpoint without ever evaluating it.

The 14 of 32 `mtcars` records reaching 20 mpg supply a transparent illustration under a separately stipulated binomial model. Counting those outcomes does not establish independent Bernoulli sampling of car models; the records and their illustrative qualification are those of Chapters 5 and 6, and no new extract or file is created.

```
nll <- function(p) -dbinom(14, 32, p, log = TRUE)
optimize(nll, c(0, 1))
14 / 32
```

Both displayed proportions are .4375 to four decimals; the small difference from exactly 14/32 is optimization tolerance. `dbinom()` includes the combinatorial term, which does not move the maximizer, and its `log = TRUE` form avoids taking a logarithm after underflow.

```
dbinom(0, 32, 0, log = TRUE)
dbinom(32, 32, 1, log = TRUE)
qlogis(c(0, 14/32, 1))
```

The three fitted probabilities are 0, .4375 and 1, with minimum negative log-likelihoods 0, 1.9588 and 0. `qlogis()` returns negative and positive infinity at the two boundaries: these are exact mathematical results, not rounded small positive probabilities, and displaying them at four decimals would misrepresent them. The intercept-only logistic fit returns $-.2513$, which is $\text{logit}(.4375)$, closing the loop between the scalar binomial case and the regression model of Section 3.

Several parameters and local curvature

Under a hypothetical independent Normal model for a quantitative outcome,

$$-\ell(\mu, \sigma) = \frac{n}{2} \log(2\pi) + n \log \sigma + \frac{\sum_i (y_i - \mu)^2}{2\sigma^2}, \quad \sigma > 0.$$

Parameterize $\sigma = e^\eta$, so an unconstrained search in η keeps the scale positive. The parameters are then (μ, η) rather than (μ, σ) , which also fixes the scale on which a numerical Hessian is reported. With positive observed variation the analytic maximizers are

$$\hat{\mu} = \bar{y}, \quad \hat{\sigma}_{\text{ML}} = \sqrt{\frac{1}{n} \sum_i (y_i - \bar{y})^2},$$

so the maximum likelihood scale divides by n where `sd()` divides by $n - 1$; this is the same distinction Chapter 9 makes before AIC and BIC. All-equal data require separate treatment, because the Normal likelihood is unbounded as σ tends to zero and no finite positive-scale maximizer exists. The `mpg` values below have positive variation. This illustration concerns a different model and a generic optimizer: it does not claim that car outcomes follow a Normal process, and it does not claim that `glm()` uses BFGS.

```
loss <- function(par) {
  -sum(dnorm(mtcars$mpg, par[1], exp(par[2]), log = TRUE))
}
start <- c(mean(mtcars$mpg) - 1, log(sd(mtcars$mpg)))
a <- optim(start, loss, method = "BFGS", hessian = TRUE)
a$convergence
eigen(a$hessian, symmetric = TRUE)$values
```

Table 17*Numerical and analytic values have different roles.*

Check	Reproduced anchor	Meaning
Analytic mean and ML scale	20.0906; 5.9320	Closed-form benchmark on the recorded data; <code>sd()</code> gives 6.0269
First-start estimates	20.0908; 5.9320	Numerical tolerance can move the last displayed digit even when convergence is reported
Both convergence codes	0	The algorithm's stopping criterion was met
First negative log-likelihood	102.3778	The same objective is compared under both starts
First minus second objective	About -5.99×10^{-7}	Near equality supports reproducibility of the numerical solution; it is not a test of the model
Hessian eigenvalues in (μ, η)	About 64.0006 and .9094	Positive local curvature of the negative log-likelihood at the returned point

A stopping code alone establishes neither agreement at the reporting precision, nor a global optimum, nor identification, nor scientific adequacy: compare starts and objective values, use an analytic benchmark where one exists, and read the returned local curvature. Section 3.2's IRLS report applies the same discipline to the fitting routine actually used, and Section 6.3 shows its symptoms when no maximum exists. **Extended practice.** Compare an interior binomial case with the zero-success and all-success cases, checking endpoint likelihoods explicitly and saying which constraint changes; for the Normal example compare the two starts with the analytic solution and state what the convergence codes and Hessian do and do not establish.

Appendix B: Changed Conditions in the Generating Process

The certification process in Section 3.3 is a reference, not a universal performance guarantee. The three changes below retain the hours coefficient .3 and alter one feature at a time. A smaller study reduces information; a lower intercept changes where observations lie on the logistic curve; a narrower range removes contrast in hours. None changes the declared coefficient being estimated.

Table 18

One changed feature per simulation condition.

Condition	Change from the baseline	Consequence to investigate
Smaller study	Reduce 1,500 students to 100	More sampling variation and greater risk of unstable estimates
Lower baseline	Set the intercept to -5	More probabilities near zero, reducing typical Bernoulli information
Narrower hours	Generate hours uniformly from 8 to 12	Less predictor spread for identifying the hours slope

For each condition, generate 500 independent datasets and retain the hours estimate, its standard error, the Wald interval and any fitting warning. The estimate distribution, its mean and SD, and coverage of .3 answer different questions: sampling stability, finite-sample bias, precision and interval calibration. Record estimates with absolute magnitude above 5 and separation warnings rather than silently deleting those replications. A warning is a screening signal, not a complete separation diagnosis.

A baseline of -5 does not make every profile's event rare: at twenty hours without tutoring the logit is 1, so the probability is .7311. Describe it as a lower-baseline process and inspect its actual prevalence. When comparing coverage across 500 repetitions, Monte Carlo variation matters: near .95 coverage its SE is about .0097. An apparent difference of a few thousandths is not compelling evidence of changed calibration.

Finally fit a linear probability model to one generated dataset from each condition and count fitted values outside $[0, 1]$. A narrower range can make the line's boundary problem less visible while making its slope less precise. Boundary protection, sampling precision and correct interpretation are distinct reasons for preferring or questioning a model.

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